

Will Joseph

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Portfolio: wrjoseph.com

EDUCATION

Mechanical Engineering B.S.

North Carolina State University

August 2022 – May 2026

GPA: 3.86

Relevant Coursework / Electives (Graduate*):

Mechanical Systems Design, Electromechanical Systems Design*, Applied Acoustics*, Control Systems for Robotics

SKILLS & SOFTWARE

Mechanical CAD / FEA: SolidWorks, PTC Creo w/ WindChill, Ansys Mechanical and Maxwell

Programming and Mechatronics: Python, UNIX, GitHub, STM32 / ESP32 / Raspi. Pi Control System Integration

Dynamic System Modelling: MATLAB / Simulink, MuJoCo, Altair MotionView and HyperStudy

Manufacturing: GD&T and Print Drawing, Manual Mill and Lathe, FDM 3D Printing, Design for Manufacturing

RELEVANT EXPERIENCE

Senior Design: Pick and Place Machine for Plant Matter (1st Place) – Raleigh, NC

October 2025 – April 2026

Team Captain

Designed and built machine to automate the transfer of plant matter between petri dishes for Syngenta scientists:

- Led early design trade-studies, final system architecture refinement, as well as mechanism design and controls integration for custom soft-touch grippers, petri dish handling, and machine vision system.
- Won 1st for achieving transfer times comparable to human, full process automation, and 98% reliability.

Constructed Facilities Laboratory – Raleigh, NC

October 2025 – May 2026

Research Assistant

Assisted in destructive testing research of full-scale reinforced concrete bridge supports. Duties include:

- Instrumentation of test structures, setup of DAQs / controllers, and setup of digital image correlation systems.
- Verification of testing procedure and data-collection.

Hybrid Intelligent Experimental Robotics Lab – Raleigh, NC

January 2025 – May 2025

Research Assistant

Self-driven research and study within the “Sim-to-Real Gap” domain in robotic controls:

- Created custom MuJoCo simulation framework for artificial training data creation and neural network training / testing.
- Combined model-based control and data-driven parameter estimation using PyTorch for parameter correction in simulation.

Volvo Trucks – Greensboro, NC

January 2024 – July 2024

Simulation Development Co-op, Steering Group

Two-rotation co-op focused on multi-body dynamic simulation development of heavy-duty truck suspensions and steering systems:

- Transferred physical test protocols for leaf springs into automated simulation and optimization software tool. Final product automatically computes non-linear kinematic parameters and principal stresses. Optimizes to desired performance targets while keeping the design space within manufacturer limits.
- Created program to accurately estimate steering wheel torque for static steering maneuvers using 1D hydraulic simulation, steering kinematics, and determination of unknown tire parameters through test data.
- Carried out several packaging studies within Creo for upcoming concept vehicles / products.

Pack Motorsports – Raleigh, NC

October 2022 – December 2023

Suspension Subsystem Member

NC State student-run Formula SAE club:

- Completed FEA analysis / report and subsequent redesign of critical suspension bearing carriers.
- Root-cause investigation of previous premature bearing failures. Analysis of assembly and dynamic loading.
- Manual and CNC manufacturing of chassis and suspension welding jigs.